

Probability

Covering

- Introduction to probability
- Relative and expected frequencies
- Probability of combined events
- Conditional probability

O levels and Extended

Extended

Probability

Probability means chance of occurrence of an event/object. Percentage of chance of occurrence written in decimal form is also called its probability. It is measured by

$$\text{Probability} = \frac{\text{Total wanted cases}}{\text{Total possible cases}}$$

Probability is between 0 and 1. Sum of all probability is 1.

Say, if probability of wanted is x , then the probability of unwanted is $1 - x$. This can be written as $P(A) = x$, $P(A') = 1 - x$.

Total possible cases

Collection of all possible outcomes of an experiment is called sample space.

e.g for tossing a coin, Total possible cases = $\{H, T\}$

e.g for rolling a die, Total possible cases = $\{1, 2, 3, 4, 5, 6\}$

Total possible cases = product of all outcome of all experiments. This is for any 2 events happening at the same time. "and" will be used to tell this.

E.g the Sample Space/Possibility diagram of Tossing a coin and throwing a die is.

Die Coin	1	2	3	4	5	6
H	$\{H, 1\}$	$\{H, 2\}$	$\{H, 3\}$	$\{H, 4\}$	$\{H, 5\}$	$\{H, 6\}$
T	$\{T, 1\}$	$\{T, 2\}$	$\{T, 3\}$	$\{T, 4\}$	$\{T, 5\}$	$\{T, 6\}$

Total possible cases = $6 \times 2 = 12$

We can find probability of 2 events by multiplying. This is for "and"

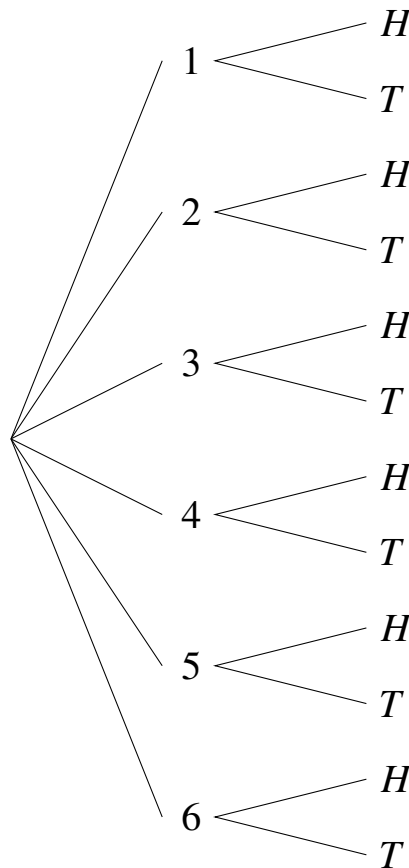
$$\text{E.g } P(\text{even}, H) = \frac{3}{6} \times \frac{1}{2} = \frac{1}{4}, \quad P(1, T) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}, \quad P(7, H) = \frac{0}{6} \times \frac{1}{2} = 0$$



Tree diagram

When you're drawing a tree diagram, from every node (branch point) you should draw as many branches as there are possible choices at that step of the experiment.

E.g for the coin and die



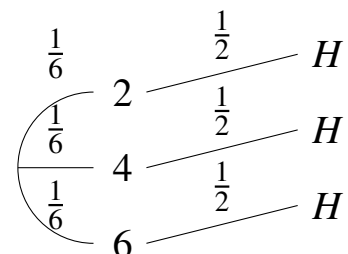
To find probability in tree diagram we write probability of each outcome of experiment on its line of tree.

At the end probabilities of the branches will be multiplied.

e.g $P(1, H)$ will be $= \frac{1}{12}$

If we have multiple branches satisfy a condition, we add.

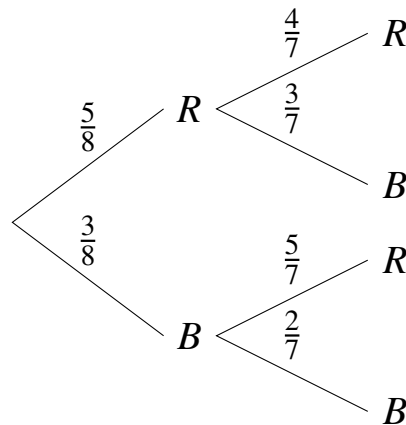
E.g $P(\text{even}, H) = 3 \left(\frac{1}{12} \right) = \frac{1}{4}$



Dependent events

Events are dependent if the result of one affects the next. This is for "without replacement" questions. This decreases the count of Total possible cases or Total wanted cases.

In a tree diagram, e.g we have 5 red balls and 3 blue balls. They are drawn without replacement, draw the tree diagram to show all possibilities



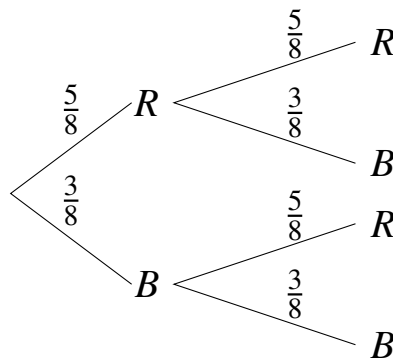
We observe that the total count of balls, denominator, and count of the ball, numerator, stays or decreases after every pick.

For these we also observe that if we have 1 branch probability missing we can find it using $1 - x$, where x is the given/found pair

Dependent cases are always with same type of object, e.g dice, cards, balls, etc.

Independent Events

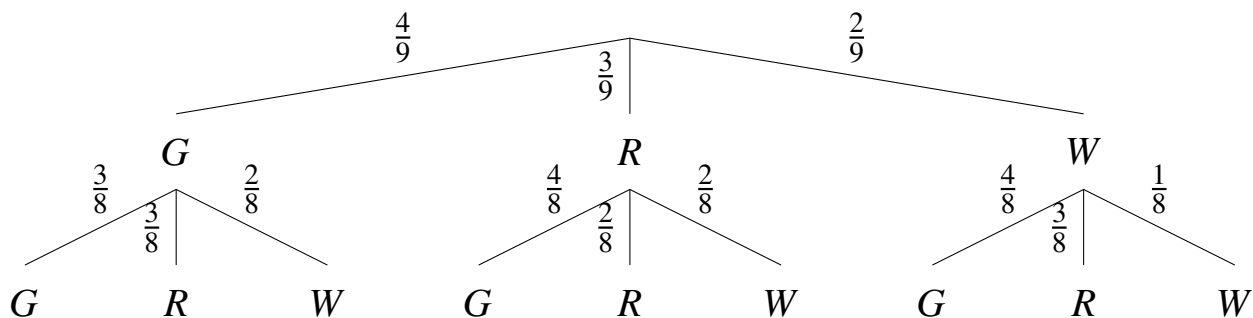
Events are independent if the result of one does not affect the other. This for the case of "with replacement" and probability of different objects. With this the tree diagram has different odds, if the balls are replaced after selection, we get:



Selection of different objects

If at a selection chooses multiple objects, we multiply possibility of individual events. If question doesn't specify order, or has "random", we multiply by number of possible orders.

E.g for a bag containing 4 green balls, 3 red balls and 2 white balls. Two balls are drawn one after another, without replacement. Find the probability of the balls being drawn in the specific order; Green, then Red



Now Green, then Red mean we take the branch $\xrightarrow{\frac{4}{9}} G \xrightarrow{\frac{2}{8}} R$, $\frac{4}{9} \times \frac{2}{8} = \frac{12}{72}$

Find the probability of drawing a Red, White.

Now this has 2 possibilities, $\xrightarrow{\frac{3}{9}} R \xrightarrow{\frac{2}{8}} W$ and $\xrightarrow{\frac{2}{9}} W \xrightarrow{\frac{3}{8}} R$. We can either add the 2 or multiply 2 by one path; $\frac{3}{9} \times \frac{2}{8} + \frac{2}{9} \times \frac{3}{8} = 2 \left(\frac{3}{9} \times \frac{2}{8} \right) = \frac{12}{72}$

Relative Frequency and Expected

O levels and Extended

Relative frequency is the probability of events relative to how many times the event happened. E.g the relative frequency of getting a 1 on a die is 0.1 after tossing it 20 times. This means in the 20 tosses, 10% of the time the die landed on 1.

We can expect a event happening for N tries using the relative frequency probability.

$E(P) = N \times P(x)$, where N is the number of tries, and P is the probability of the event in the relative frequency table.



Question

Explanation has cases for 2 probabilities, this works for 3 and more also.

When we have "and" in a question, we multiply probabilities. e.g Find probability of $P(a)$ **and** $P(b)$ happening, the answer will be $P(a) \times P(b)$.

E.g find the probability of landing on yellow and red on the respective spinners. $P(\text{yellow, red}) = P(\text{yellow}) \times P(\text{red})$.

When we have "or" in a question, we add probabilities. e.g Find probability of $P(a)$ **or** $P(b)$ happening, the answer will be $P(a) + P(b)$.

E.g find the probability of choosing White and Blue cars. $P(\text{White, Blue})$ or $P(\text{Blue, White})$, this or is why we multiply number of possible orders. $P(W, B) + P(B, W) = 2 \times P(W, B)$

Conditional Probability

Extended

Conditional probability measures the likelihood of an event A occurring given that another event B has already occurred.

E.g A group of 200 people were asked which city they would like to visit next. The table shows the results.

City	London	Paris	New York	Tokyo
Number of people	50	48	56	46

Two people are chosen at random from the group of 200.

Find the probability that one person would like to visit London next and the other person would like to visit New York next.

Give your answer as a percentage.

$$P(\text{London}) = \frac{50}{200}$$

Now since 1 person has picked, 199 have yet to choose

$$P(\text{New York}) = \frac{56}{199}$$

Since we could have done New York $\rightarrow$ London also, we $\times 2$

$$\frac{50}{200} \times \frac{56}{199} \times 2 = \frac{29}{199}$$

$$\frac{29}{199} \times 100\% = 14.573\% \text{ (3 d.p.)}$$



Conditional Probability

E.g Sonia has a bag containing 20 sweets. In the bag, there are 5 red, 6 green and 9 yellow sweets.

Sonia chooses two sweets at random, without replacement, from the bag.

(i) Find the probability that she chooses two green sweets.

$$P_1(\text{Green}) = \frac{6}{20}$$

Now since 1 green is picked, there are 19 total sweets and 5 green sweets.

$$P_2(\text{Green}) = \frac{5}{19}$$

Now we combine

$$P(\text{Green and Green}) = \frac{6}{20} \times \frac{5}{19} = \frac{3}{38}$$

(ii) Calculate the probability that the sweets she chooses are **not** both the same colour.

We see the possible cases are:

Red $\rightarrow$ Green, Red $\rightarrow$ Yellow and Yellow $\rightarrow$ Green.

$$P(R \rightarrow G) = \frac{5}{20} \times \frac{6}{19} \times 2 = \frac{3}{19}$$

$$P(R \rightarrow Y) = \frac{5}{20} \times \frac{9}{19} \times 2 = \frac{9}{38}$$

$$P(G \rightarrow Y) = \frac{6}{20} \times \frac{9}{19} \times 2 = \frac{27}{95}$$

Their sum is the final answer

$$\frac{3}{19} + \frac{9}{38} + \frac{27}{95} = \frac{129}{190}$$

